CHARUSAT Online Course Chatbot

Complete Engineering Audit: All Development Rounds (R1 to Latest Before vs. After)

Project Evolution Overview: This audit clarifies all changes made across the 5 development phases of the CHARUSAT RAG Chatbot. It traces the journey from an initial single-file script to an enterprise-grade academic assistant with verified hybrid retrieval, multi-turn dialogue memory, dynamic typing autocomplete, and a 5-tab admin portal.

1. Detailed Chronological Breakdown: What Happened in Each Round

Round / Phase	BEFORE (Initial Approach & Problems Faced)	AFTER (Fix Implemented & Latest Result)
Round 1: Scraper & CLI Prototype	• Crawled *charusat.online*; parsed PDFs with PyMuPDF. • Built terminal CLI (*main.py*). **Problems:** Free Groq API rate limits (429); single-model crashes; noisy markdown chunks; no web UI.	• Implemented polite 0.5s crawling delay (max 250 pages). • Designed multi-model failover architecture. • Filtered chunks < 50 chars and exact duplicates. • Prototyped basic RAG chain successfully.
Round 2: Web App & Cloud Deploy	• Created modern React + Tailwind frontend. • Built FastAPI HTTP server (*api.py*). • Added MongoDB authentication & chat history. **Problems:** Render build path confusion; cold starts on free tier; Vercel preset mismatch.	• Created standardized *render.yaml* and *build.sh* for backend. • Configured *vercel.json* for static SPA frontend. • Created dedicated *deploy* branch for isolated hosting. • Added Server-Sent Events (SSE) streaming API.
Round 3: Quality Hardening (Steps 1–11)	• Ran 40-question and 10-core diagnostic benchmarks. **Problems:** Score threshold bug (RRF compressed scores to 0.015–0.033 below 0.52 gate, causing 19/40 questions to return 0 chunks); BGE cross-encoder reranker hurt refund policy accuracy; MCA chunks leaked into BCA queries; regex text cleaning broke sentences.	• Switched to Natural Cosine Similarity (0.0–1.0) with calibrated 0.35 threshold (100% answerable queries pass). • Added BM25 metadata filtering for strict program isolation. • Disabled cross-encoder reranker (improved policy retrieval). • PyMuPDF multi-row header merging for PDF tables. • Integrated *remark-breaks*; benchmarked 1.13s latency.
Round 4: UI Overhaul & Admin Portal (Steps 0–8)	• Baseline snapshot (tag *backup-before-round4*). **Problems:** Desktop sidebar toggle bug locked sidebar closed; source chips cluttered chat bubbles; bare follow-ups (e.g. 'mca' after 'MBA fee?') lost topic; erratic currency (*INR 50,000*, ■ *50000*); unverified landing page stats; no admin management.	• Step 1: Removed public source chips (kept in audit logs). • Step 2: Built dynamic typing autocomplete from 117-question bank + *GET /api/contacts*. • Step 3: Fixed desktop sidebar with *localStorage* state. • Step 4: Refactored *Chat.jsx* into *useChat* & *useSidebar* hooks. • Step 5: Added regex currency normalizer (■*50,000*) and multi-turn topic retention. • Step 6: Overhauled landing page with 4 degree cards and topic badges. • Step 7: Built 5-Tab Admin Dashboard with JWT auth and rate limiting.
Latest Work: Presentation & Report Audit	• Final-year AI/ML student preparing for mentor presentation. **Problems:** Old reports had stale claims (claimed 444 chunks vs actual 345 on main / 389 on deploy; claimed active reranker).	• Computed every metric directly from code, database, and git. • Generated verified 7-page report (*CHARUSAT_Chatbot_Journey_Report.docx*). • Built compact architecture flow diagram and mentor Q&A; guide.

2. Core System Matrix: Before vs. After Upgrades

Feature / Component	BEFORE (Original State / Flaws)	AFTER (Latest Upgraded State)
Chat UI & Navigation	• Monolithic 530-line component. • Desktop toggle button triggered mobile drawer, locking sidebar shut. • Cluttered source chips under every bubble.	• Modular custom hooks (*useChat.js*, *useSidebar.js*). • Desktop sidebar persistent in *localStorage*. • Clean bubbles with source chips preserved only in backend logs.
Typing Autocomplete	• No typing autocomplete. • Static starter cards. • Students had to guess what questions the bot could answer.	• Dynamic keyboard-navigable autocomplete suggestions. • Powered by verified 117-question bank (92 Full, 21 Partial, 4 Not-in-Docs).
Answer Formatting & Quality	• Markdown labels ran together (***Degree:**Online MBA**Mode:***). • Erratic currency notation (■*50,000 INR*, *Rs. 50,000*). • Context lost on bare programme follow-ups.	• Integrated *remark-breaks*; enforced structured comparison tables. • Automated regex currency normalizer (■*50,000*). • Topic retention preserves topic/year across follow-ups.
RAG Retrieval Engine	• Artificial RRF score gate dropped 19/40 questions. • Cross-encoder reranker hurt institutional policy retrieval. • Fragmented table chunking.	• Balanced Hybrid Search (Dense BGE-Small 384d + Sparse BM25 via RRF k=60). • Threshold calibrated at 0.35. • 345 clean chunks with 3,600-char table chunking in PyMuPDF.
LLM Reliability & Fallbacks	• Single model dependency. • Free tier 429 rate limit or timeout crashed entire system.	• Resilient 3-tier fallback chain: 1. *llama-3.3-70b-versatile* → 2. *llama-3.1-8b-instant* → 3. *gemini-1.5-flash* with auto circuit breaker.
Public Portal & Branding	• Generic landing page with unverified claims (*'10k+ students'*) and static dead links.	• Institutional CHARUSAT portal: 4 degree cards, 6 topic discovery badges, 3-step student journey, and dynamic *GET /api/contacts*.
Admin Management & Security	• No admin portal, no feedback capture, no rate limiting, no error audit logs.	• 5-Tab Admin Dashboard (Overview, Chat Logs, Question Bank, Feedback, Settings) with JWT auth and thumbs up/down tracking.
Git & Deployment State	• Deploy branch was 21 commits behind main. • Vector DB discrepancies (389 vs 345 chunks).	• Production requirements isolated in *backend/requirements.txt*. • Documented Render/Vercel pipelines and 21-commit delta.

3. Latest Verified Numbers & Operational Parameters

Operational Parameter	Verified Value	Technical Authority / Code Source
Vector Database Chunks (main)	**345 chunks** (198 Web Markdown, 147 PDF)	ChromaDB at *backend/vector_db*; min: 89, avg: 645.5, max: 3,567 chars.
Cloud Vector DB (deploy branch)	**389 chunks** (237 Web Markdown, 152 PDF)	Deployment branch DB prior to Round 3/4 clean rebuild; 21 commits behind main.
Embedding Model & Dimensions	**BAAI/bge-small-en-v1.5** (384 dimensions)	FastEmbed in *backend/app/embeddings.py*; squared L2 converted to Cosine.
Question Bank Classification	**117 total**: 92 Full, 21 Partial, 4 Not-in-Docs	*backend/app/data/question_bank.json*; verified against official university documents.
Retrieval Threshold & Max Chunks	Score >= **0.35**; Top **5 chunks** to LLM	*backend/app/config.py* (*SCORE_THRESHOLD = 0.35*, *MAX_CHUNKS = 5*).
Average Response Latency	**1.13 seconds** (0.81s – 2.01s range)	Verified from *CHANGES/round3_logs.json* across 10 core diagnostic questions.